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RESEARCH ARTICLE

Teach Pendant at Fingertips: Intuitive Vision-Based Gesture-Driven Control of Dexter ER2 Robotic Arm

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ABSTRACT Teaching a robotic arm can be unintuitive and challenging due to the disparity between natural human limb movements and robotic manipulator control. Substantial training is often required to effectively operate such systems. This work presents the development of an intuitive, easy-to-use interface that maps human palm gestures to control the movement of a 5-axis robot, the Dexter ER2. The proposed vision-based system enables the robotic arms to autonomously follow hand gestures through a three-stage methodology: gesture detection, gesture tracking, and robotic manipulator control. A deep learning-based model is employed for accurate gesture recognition, allowing the robot to mimic the controller's limb movements with minimal cognitive load. For trajectory computation and joint movement generation, an adaptive gradient descent-based inverse kinematic solver is implemented, enabling efficient and smooth convergence of joint angles corresponding to the desired end-effector positions. The interface was successfully implemented, enabling the robot to follow gesture commands and reach target positions. The proposed vision-based gesture-driven control scheme of the Dexter ER2 robot arm is validated through both simulation and experimental studies. The system employed serial communication, ensuring a continuous and lossless data stream between the controller and the computer. A detailed comparative analysis is also presented to highlight the advantages of the proposed approach. Future work includes enhancing visual appeal with a graphical interface and addressing the latency issue in real-time control.

INDEX TERMS Cognition, gesture, deep learning, object detection, object tracking, 6-DOF robotic arm.

I. INTRODUCTION

A robotic arm is a technological mimic of the human arm. Robotics has made its way into many industries, and transformed how tasks are performed. While the control of robotic arms represents a significant advancement, modern robotic manipulators are now operated and programmed with greater flexibility. A wide variety of such arms can be used. Historically, the primary means of controlling a robotic arm included a joystick, a keyboard, or a specialized control panel. However, these methods are often unintuitive and may require great deal of practice to achieve proficiency [1].

Cognition refers to the psychology of human thinking and the mental processes of getting information from the world. It is an orderly and logical approach to problem solving, gathering information, and making decisions. Analytic cognition requires deliberate effort and is limited by the amount of readily available data. In contrast, intuitive cognition involves unconscious situational pattern recognition and synthesis, which are not limited by working memory and allow for faster responses. It is based on the experience towards the object of activity. The literature on naturalistic decision making comprises a recognition-primed decision model in which

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intuitive cognition plays a central role. This model suggests that analytical cognition is a type of intellectual exoskeleton that offers extra capabilities, such as working memory, while intuitive cognition serves as the foundation for situational understanding and meaning making.

All man-made technologies are just an extension of one's capability or to assist the disabled [2]. Hence, these must be easy and intuitive to use. The gap in machine control and intuitive control of the body limb demands some analysis as to which control key leads to what effect. This is usually experienced by beginners in simulated games, where there is a complex set of controls for various actions to be performed in the gaming environment. A classic experience of steering a cycle backward would need some adjustments, as it is intuitively opposite to that in the forward direction [3]. A bit complex example in the robotics field would be the teleoperation of TurtleBot [4]. Since turtlebot is a nonholonomic robot, using control to different directions is not only redundant but also makes it complex for the operator, especially to change speeds and to fall back to the same speed on running again. The teleoperation designed for the turtlebot is much intuitive [5].

Such a situation is also encountered while operating a multi-degree-of-freedom (DOF) robotic arm, where each joint is to be actuated by a different control. Although a robotic arm mimics the structure of a human limb than a mobile robot, the control is still much less intuitive than the simpler control of planar mobile robots. In the present field of study, the goal is to develop an intuitive, user-friendly control mechanism that is cost-effective and takes less computational time by leveraging the natural motion of the human arm. Also, to simplify the control process, researchers have done various gesture-based control [6], [7], [8], [9] as one of the control strategies for robotic arms. Using hand motions, users can interact with the robotic arm through hand pose estimated control, which can be more simpler method to learn than conventional control techniques [6], [8], [9]. This method can improve robotic arm accessibility and usability, making them more adaptable and useful for a wider range of applications [6], [7], [9].

The primary focus of this presented research work is applied rather than theoretical. The present work emphasizes the integration of techniques such as anthropomorphic control, gradient descent with adaptive learning rates for inverse kinematics (IK), and single-camera pose estimation into a unified vision-based control system. The novelty lies in the practical application of these techniques, aiming to develop an intuitive, cost-effective interface for controlling robotic manipulators in constrained environments. By focusing on usability, affordability, and adaptability, this study bridges the gap between theoretical robotics and its practical implementation.

Researchers have been exploring the concept of gesture-controlled and pose-estimated controlled robotics, going through various techniques and technologies to capture and process human hand gestures or various pose landmarks. One common approach is using cameras to detect user hand movements and send as control commands for the robotic arm; this can be termed as Anthropomorphic control [10]. Anthropomorphic control of a manipulator refers to controlling a robotic arm or manipulator in a way that mimics or maps to the natural motion of human hand, i.e. it involves designing the manipulator system such that the operator can intuitively control the motion of the end-effector by making arm or hand motions or gestures that the manipulator will follow [11]. As research in this field continues to advance, we can expect to see more innovative applications of robotics in the future.

II. RELATED WORK

In recent years, with advancements in deep learning and camera technologies, vision-based robot control has seen a rapid growth, in both theoretical foundations and innovative applications. This brief review focuses on a few latest developments and their applications, particularly in the context of robot control using gestures.

Gupta et al. [7] proposed a continuous hand gesture recognition technique for human-machine interaction using a gyroscope sensor and accelerometer, demonstrating high classification accuracy and real-time responsiveness. However, their method relies on the wearable device, limiting natural user interaction and requiring sensor calibration, which restricts its intuitiveness and scalability in the vision-based robotic control system. Atre et al. [8] developed a low-cost, rule-based robotic arm control system using a webcam, Arduino, and Bluetooth module. Tested in simulation, the system was limited by low precision, restricted DOF, and lacked AI-integration or real-time control.

In addition to this, Angulo et al. [12] demonstrated the use of stereo vision for robotic arm control in a critical application, such as positioning near an explosive device. This highlights the importance of vision-based control in hazardous environments. Solly and Aldabbagh [9] implemented a wearable hand glove controller system for a mobile robot using flex sensors, an accelerometer, a Bluetooth module, and a radio module. The gesture mapping approach enabled robot navigation but lacked suitability for precise manipulation tasks. The system was experimentally validated and involved a high cost due to the wearable hardware.

Further, Altayeb [13] developed a robotic arm finger control system using a webcam for simple vision thresholding, along with Arduino, a servomotor, and a buck converter. The system was experimentally validated and offered a moderately priced solution, but faced limitation in scalability. Qin et al. [14] utilized a monocular RGB camera and imitation learning algorithm to enable real-time gesture-based teleportation of multi-fingered dexterous robots. The system demonstrated high potential but faced challenges such as loss of tracking during fast motion and unreliable hand pose estimation, with a relatively high implementation cost.

Furthermore, Terreran et al. [15] presented a contactless interface for collaborative robots, achieving promising results with natural human-robot interaction. However, limitations remain in achieving precises manipulation and robustness under varying lighting and background conditions, pointing to the need for improved visual processing and real-time gesture interpretation. Suhaimi et al. [16] presented a control system for a 4-DOF robotic arm using Electrooculogram (EOG) and Electromyography (EMG) sensors, combining EMG-based gaze estimation with image processing techniques. The system involves complex sensor fusion and incurred a very high cost.

Moreover, Godoy et al. [17] proposed an affordable and EMG-based tele-manipulation framework for intuitive control of robotic arm-hand systems, demonstrating improved task execution in dynamic environments through multimodal sensing. However, the reliance on EMG signals and predefined object affordances limits adaptability and user generalization, highlighting the need for more natural and flexible interaction methods such as vision-based gesture control. Ban et al. [18] proposed a persistent human machine interface for robotic arm control using gaze and eye gestures, demonstrating high accuracy and minimal calibration for disabled users. However, the approach requires specialized eye-tracking hardware, limiting accessibility and scalability for border-intuitive control systems.

Additionally, Bamani et al. [19] proposed a gesture recognition system for the quadruped robot using an RGB webcam and a super-resolution vision model. The system was experimentally validated but faced limitations in handling a restricted range of gestures and performing challenges under varying environmental conditions. Angelidis and Bampis [20] present a gesture-controlled robotic arm system for small assembly lines, highlighting its effectiveness in intuitive human-robot interaction using AI and RGB-D sensors. However, the system's precision is limited by the depth camera accuracy, particularly in estimating small surfaces link fingertips.

Similarly, Ferin et al. [21] demonstrated the effectiveness of using computer vision techniques to enable intuitive and contactless human-robot interaction. However, limitations include the sensitivity to lighting conditions and occlusions, which can affect gesture recognition accuracy and system reliability. Assis et al. [22] explores machine learning (ML) based control for a 6-DOF robotic arm, utilizing computer vision to capture operator movements and train a neural-network for predicting joint angles. While the approach demonstrates promising accuracy and adaptability, limitations include sensitivity to noise in depth estimation, which affects system performance.

Although gesture-based robotic control has been widely investigated, many approaches still face key limitations in deployment. Wearable sensor systems, such as those by [7], [8], and [9], demonstrate high responsiveness but restrict natural interaction and scalability due to reliance on external hardware. Vision-based methods using MediaPipe or stereo cameras [12] offer a more intuitive interface but are often limited to simple commands or controlled environments. Systems like those proposed in [13] and [14] show promise with monocular cameras and deep learning but are constrained by predefined gestures, high computational demands, or reduced robustness in dynamic settings. Similarly, while multimodal and depth-based systems [16], [20] enhance recognition accuracy, their dependence on specialized hardware impedes broader accessibility. Recent works integrating gaze [18], EMG [17], or RGB-only models [19], [21] show evolving capabilities but often lack real-time arm manipulation.

Thus, there remains a gap in developing a low-cost, intuitive, and vision-only gesture-driven robotic control system that enables accurate, real-time, and fine-grained manipulation. This motivates the proposed integration of single-camera-based deep learning gesture recognition with responsive control of a 6-DOF robotic arm using optimized gradient descent and experimental validation.

The objective of the present work is to develop a low-cost, vision-based interaction system that enables intuitive and precise control of the 5-axis Dexter ER2 robotic arm through natural palm gestures. By eliminating the need for wearable sensors or specialized hardware, the system aims to bridge the gap between usability and complexity, minimizing training overhead while enhancing accessibility and real-time responsiveness for practical applications. Moreover, the highlights of the presented work as a prime contribution are as follows:

- The proposed control framework incorporates a practical optimization approach by introducing a gradient descent-based IK solver with multiple adaptive learning rates, ensuring responsive and precise control of a 6-DOF robotic arm while emphasizing usability, cost-effectiveness, and operational safety for practical deployment.
- A vision-based control system is presented that enables intuitive and practical manipulation of the robotic arm by mapping human limb gestures to control robot motion using single camera pose estimation by employing deep learning-based object detection and tracking.
- The complete system, including hardware-software integration, is validated through both simulation and real-time experiments, demonstrating the reliability, effectiveness, and feasibility of the presented approach.

The rest of the paper is organized as follows. In Section III-A, the kinematic modeling of the Dexter ER2 robot is described, and Section III-B presents an adaptive gradient descent-based IK solver. Then, the selection of visual features for efficient tracking of moving objects is presented in Section IV, in which visual features for tracking control of 6-DOF (Section IV-A) are presented, followed by a discussion of gesture recognition framework (Section IV-B) and recognition to actuation angles (Section IV-C). Section V presents the development of a proposed vision-based controller to track the moving object. Section VI depicts the results, in which employed software and hardware components (Section VI-A) are described, followed by comprehensive simulation results (Section VI-B). Further, detailed analysis on experimental results are presented in Section VI-C. Furthermore, robustness analysis is demonstrated in Section VII. Section VIII presents a comparative analysis. Detailed discussion is given in Section IX. The conclusion and scope of further work are given in Section X.

III. KINEMATIC MODELLING OF DEXTER ER2, A 6-DOF ROBOTIC ARM

A. DYNAMICS OF 6-DOF ROBOTIC ARM

The robotic manipulator used for our experiment is a 6-DOF manipulator, featuring six joints allowing for a wide range of motion. The manipulator consists of six revolute joints, as depicted in Figure 1.

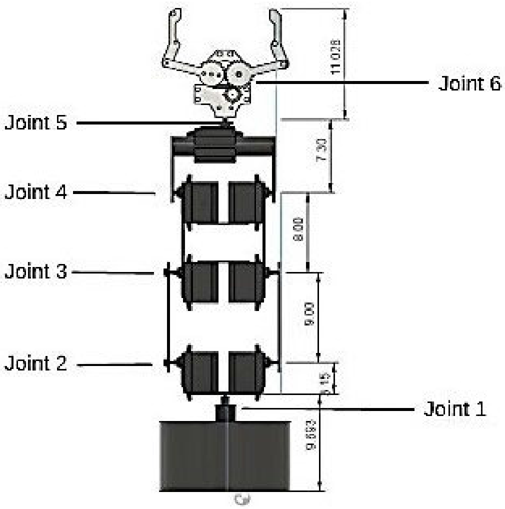


FIGURE 1. Joints and links in Dexter ER2.

Each joint rotates along a specific axis, contributing to the overall spatial movement of the manipulator. The length of each link in the manipulator arm is provided in the Table 1 [23].

TABLE 1. Lengths of links in Dexter ER2.

Links	Length
Link1	9.6 cm
Link2	3.1cm
Link3	9 cm
Link4	8 cm
Link5	7.3 cm
Link6	11.0 cm

Kinematic analysis and modeling of the manipulator often rely on Denavit-Hartenberg (DH) parameters, commonly known as DH parameters. These parameters form the basis for describing the kinematic relationships between adjacent links and joints in the manipulator's kinematic chain. Figure 2 illustrates the DH model representation of the 6-DOF robotic manipulator, providing a visual depiction of the spatial arrangement of joints and links. Utilizing this model, the kinematics of the manipulator can be accurately described and analyzed. Kinematic modeling plays a crucial role in predicting the behavior of the manipulator during operation.

TABLE 2. DH parameters for Dexter ER2.

$Link_i$	θ_i	D_i	a_i	α_i
1	0	L_1	0	0
2	θ_1	L_2	0	90
3	θ_2 + 90	0	L_3	0
4	θ_3	0	L_4	0
5	θ_4 - 90	0	0	-90
6	0	L_5	0	0
7	θ_5	L_6	0	0

Once the DH parameters are determined for five revolute joints RRRRR with joint variables θ_1, θ_2, θ_3, θ_4, θ_5, the DH model is developed as illustrated in Figure 2. The DH parameters are tabulated in Table 2. After obtaining the DH parameters, the problem simplifies significantly. Utilizing the DH matrix, a transformation matrix for each frame can be derived as

$$T_0 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

$$T_1 = \begin{bmatrix} \cos\theta_1 & 0 & \sin\theta_1 & 0 \\ \sin\theta_1 & 0 & -\cos\theta_1 & 0 \\ 0 & 1 & 0 & l_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

$$T_2 = \begin{bmatrix} \cos(\theta_2+90) & -sin(\theta_2+90) & 0 & l_3\cos(\theta_2+90) \\ sin(\theta_2+90) & \cos(\theta_2+90) & 0 & l_3 sin(\theta_2+90) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

$$T_3 = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & l_4\cos\theta_3 \\ \sin\theta_3 & \cos\theta_3 & 0 & l_4\sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

$$T_4 = \begin{bmatrix} \cos(\theta_4-90) & 0 & -sin(\theta_4-90) & 0 \\ sin(\theta_4-90) & 0 & \cos(\theta_4-90) & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{5}$$

$$T_5 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & l_5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

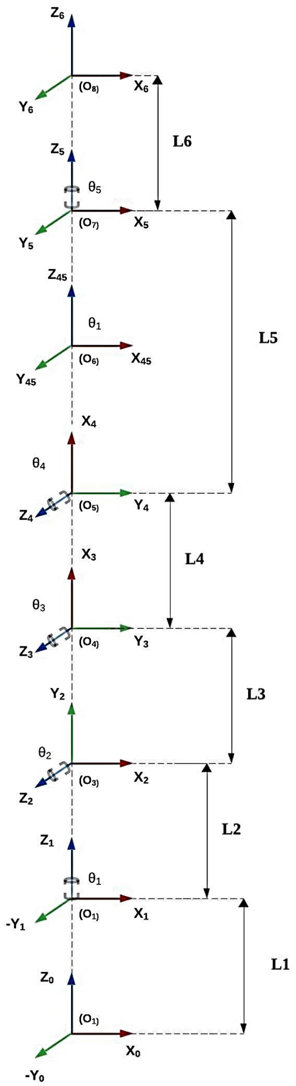


FIGURE 2. **DH model representation of Dexter ER2.**

$$T_6 = \begin{bmatrix} \cos\theta_5 & -\sin\theta_5 & 0 & 0 \\ \sin\theta_5 & \cos\theta_5 & 0 & 0 \\ 0 & 0 & 1 & l_6 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{7}$$

Subsequently, the forward kinematics equation (or mathematical representation of the model), namely the final transformation matrix representing the pose of the third frame with respect to the zeroth frame, is obtained. By employing forward kinematics equations (1) to (7), the position of the end effector relative to the world frame can be determined, given a specific set of joint angles or rotations. Cartesian coordinates of the end-effector pose along the x, y, and z axes can be calculated using the equations (8), (9), (10), as shown at the bottom of the next page. The partial derivatives of the equations (8), (9), (10) are used to solve for IK.

Algorithm 1 Adaptive Gradient Descent-Based Inverse Kinematics for 6-DOF Robot Arm

Require: Desired end-effector pose $\mathbf{P}_{target}$, initial joint angles $\theta_0 = [\theta_1, \theta_2, \theta_3, \theta_4, \theta_5]$, learning rates α_i, tolerance ϵ, maximum iterations N_{max}

Ensure: Joint angles θ that achieve target pose

Initialize $\theta \leftarrow \theta_0$
for $n = 1$ to N_{max} **do**
  Compute current pose $\mathbf{P}_{current}$ using forward kinematics
  Compute error: $F = \|\mathbf{P}_{target} - \mathbf{P}_{current}\|^2$
  if $F < \epsilon$ **then**
    break {Convergence achieved}
  end if
  for each joint angle θ_i **do**
    Compute partial derivative $\frac{\partial F}{\partial \theta_i}$ using symbolic differentiation
    Update $\theta_i \leftarrow \theta_i - \alpha_i \cdot \frac{\partial F}{\partial \theta_i}$
  end for
end for
return Optimized joint angles θ

B. ADAPTIVE GRADIENT DESCENT-BASED INVERSE KINEMATICS FOR 6-DOF ROBOT ARM

IK involves computing joint angles of a robotic manipulator based on a desired end-effector pose (position and orientation). For a 6-DOF robotic manipulator, this translates into determining six joint variables that achieve a specific spatial configuration. Due to the inherent non-linearity of robot kinematic equations, obtaining an analytical solution can be infeasible or computationally intensive. Gradient decent is a widely used IK method in robotics due to its simplicity and robustness. Unlike Jacobian-based approaches, it avoids complex matrix computations and handles singularities more gracefully. Compared to neural network methods, it does not require large training datasets and offers greater transparency. These advantages make it a practical choice for present robotic application.

Gradient descent is an iterative optimization algorithm that is used to minimize a loss function. It works by iteratively updating the parameters (joint angles) in the direction that reduces the error most effectively. At each iteration, the gradient of the objective function with respect to the joint angles is computed. The gradient indicates the direction of steepest ascent in the parameter space. By moving in the opposite direction of the gradient, the joint angles are updated to reduce the error between the desired and actual poses. The step size or learning rate controls the magnitude of each update, balancing convergence speed and stability. A small learning rate ensures convergence but may lead to slow convergence, while a large learning rate may cause oscillations or divergence.

In this implementation, an objective function F is defined that calculates the squared Euclidean distance between the current end-effector pose and the desired target pose. This distance represents the "error" that the optimization tries to minimize. The partial derivatives of F with respect to each joint angle $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$ are computed symbolically. These derivatives represent the gradient of the function, indicating the direction of steepest ascent (increasing error). This gradient guides the iterative adjustment of joint angles to minimize the pose error.

The gradient descent used to solve for IK is presented as a flowchart in Figure 3 and Algorithm 1:

Python uses *SymPy* to symbolize the kinematic expressions for the end-effector coordinates (X_R, Y_R, Z_R) in terms of joint angles $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$. This avoids hard-coding specific values for arm lengths and target coordinates, making the code more flexible.

IV. FEATURE SELECTION

A. VISUAL FEATURES TO CONTROL 6-DOF

To train the robot intuitively, certain features of the trainer are selected, so that the training process becomes a layman's task. A robot manipulator is just a technological extension of human arm, hence, using the parameters of the trainer's arm should be the best choice. However, moving the entire arm is much effort, and getting the whole scene into the training visual frame would be a challenge with the existing miniature cameras. Hence, the index finger, being the most dexterous and tangible, is used to control the basic joints of the robot. This is the most intuitive part of control, where the robotic arm mimics the arm movement in general. Rest of the joints are actuated through different gestures, such as the pitch angle of the end effector controlled by

$$
\begin{aligned}
X = {} & L_3 cos\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) - L_6\left[cos\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& + cos\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) + sin\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\\
& \left.- cos\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right] + L_5\left[\cos\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& + cos\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) + \sin\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\\
& \left.- cos\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right] + L_4\left[cos\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& \left.- cos\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right]
\end{aligned} \tag{8}
$$

$$
\begin{aligned}
Y = {} & L_3 \sin\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) - L_6\left[cos\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& + sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) + sin\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\\
& \left.- sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right] + L_5\left[\cos\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& + sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) + \sin\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\\
& \left.-sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right] + L_4\left[cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\theta_1}{180}\right) cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& \left.- sin\left(\frac{\pi\theta_1}{180}\right) sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right]
\end{aligned} \tag{9}
$$

$$
\begin{aligned}
Z = {} & L_1 + L_2 + L_3 \sin\left(\frac{\pi\theta_2}{180}\right) + L_5 * \left[\cos\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) \cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) - sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\theta_2}{180}\right)\right.\\
& \left.- sin\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right) + sin\left(\frac{\pi\theta_3}{180}\right) \cos\left(\frac{\pi\theta_2}{180}\right)\right]\\
& + L_6\left[\cos\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) \cos\left(\frac{\pi\,(\theta_2+90)}{180}\right) - sin\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right.\\
& \left.- \sin\left(\frac{\pi\,(\theta_4-90)}{180}\right) cos\left(\frac{\pi\theta_3}{180}\right) sin\left(\frac{\pi\,(\theta_2+90)}{180}\right) + sin\left(\frac{\pi\theta_3}{180}\right) \cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right]\\
& + L_4\left[cos\left(\frac{\pi\theta_3}{180}\right) \sin\left(\frac{\pi\,(\theta_2+90)}{180}\right) + sin\left(\frac{\pi\theta_3}{180}\right) \cos\left(\frac{\pi\,(\theta_2+90)}{180}\right)\right]
\end{aligned} \tag{10}
$$

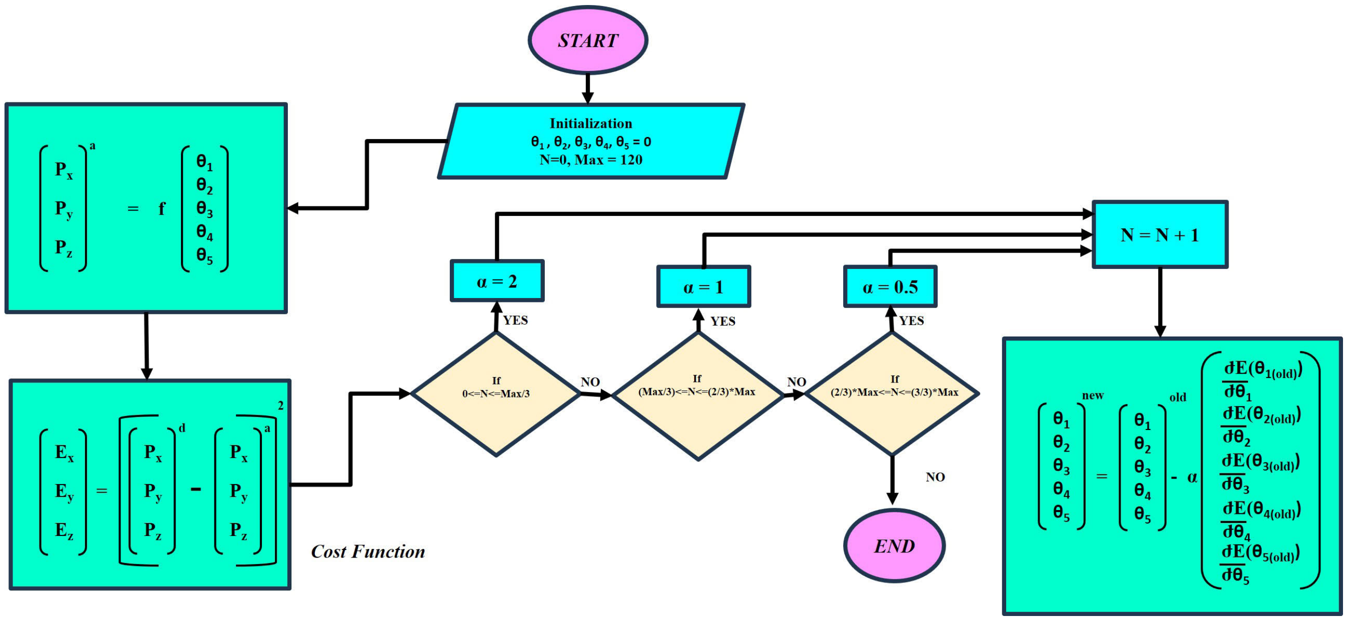


FIGURE 3. Adaptive gradient descent flow chart.

movement over a conventional cricket bowling pitching gesture.

B. GESTURE RECOGNITION FRAMEWORK

MediaPipe is an efficient deep learning framework used to process time-series data. It is open-source, lightweight, modular, and customizable. One of the algorithms that MediaPipe provides is hand detection. It produces a perception pipeline called a graph consisting of nodes called calculators, at various joints and knuckles, connected by edges called streams. By feeding a stream of images into the MediaPipe Graph, hand landmarks and their position in the image can be generated. The algorithm has the following workflow:

1) Image Input: Image typically captured by a camera or video source containing a hand is input.
2) Pre-processing: The input image is processed to reduce noise and improve quality. This includes resizing of image, converting it to grayscale, and applying a Gaussian blur filter.
3) Hand Detection: The pre-processed image is passed through a hand detection model based on Single Shot Detector (SSD) architecture with a MobileNetV2 backbone identifies and localize the hand region in the input image by regressing a rotated bounding box around the hand. The model is trained over a big datasets of hand images in different gestures and situations [24].
4) Hand Landmark Estimation: Once the hand is detected, the region of interest is cropped, aligned, and passed to a lightweight, custom convolutional neural network (CNN) designed for hand landmark estimation. This second-stage model predicts the 3D coordinates of 21 key joints on the palm which includes knuckles and fingertips with high spatial precision., which includes knuckles and fingertips.
5) Hand-Robot Tracking: The combination of efficient detection and accurate landmark localization enables robust real-time hand pose tracking using only a single RGB camera. The estimated hand landmarks are then used to track the hand as it moves in the video or image frames. It uses a combination of Kalman filters and optical flow to predict the hand's position and track its movement over time, and these tracked positions are subsequently used to extract gesture-based control for the Dexter ER2 robotic arm in the proposed system.

FIGURE 4. Gesture recognition framework.

Figure 4 shows the gesture recognition framework. The hand detection module of Figure 4 provides coordinates of the 21 landmarks on the palm, in the image. Since the image is to be taken from a stationary point of view and a stable platform, there would be no need for any further transformations. The palm is to be placed orthogonal to the camera vector to access maximum range for the limited range of a finger's movements.

C. RECOGNITION TO ACTUATION ANGLES

Image coordinates of 21 different landmarks of the palm provided by MediaPipe greatly serve to estimate the gesture and derive the corresponding angles to actuate the motors of the robotic arm. Each gesture is closely examined to determine the horizontal and vertical limitations of a palm landmark in a particular gesture, such that they are mutually exclusive.

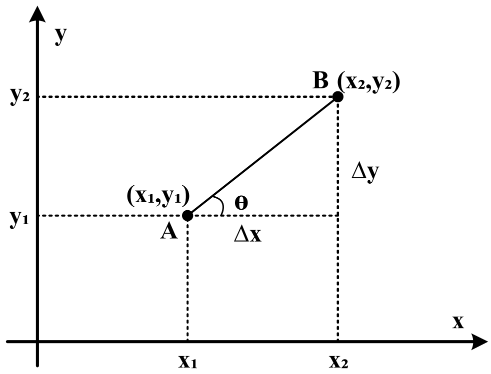


FIGURE 5. Horizontal angle of line connecting two coordinates.

From Figure 5, which depicts the angle made by the horizon with the line connecting 2 points on a plane

$$tan\theta = \Delta y \, / \, \Delta x \tag{11}$$

$$\theta = tan^{-1} (\Delta y \, / \Delta x) = tan^{-1} (y_2 - y_1 / x_2 - x_1) \tag{12}$$

These inequalities are conditioned into the code to detect the gesture. Then the desired angle between different knuckles is found using trigonometry. This enables efficient gesture-to-motion mapping with low-latency and robustness performance across diverse environment, without relying on heavy computation or training. The tangent function for an angle is defined as the ratio of the distance of the side opposite and the adjacent to the angle, other than the hypotenuse. For the angle between any two arbitrary knuckles with the horizon, the inverse of this tangent function is applied over this ratio. Corresponding pulse signal controls are delivered to the appropriate actuating motors through motor drives.

V. PROPOSED VISION-BASED CONTROL SYSTEM

The system follows a step-by-step process, as illustrated in Figure 6. Initially, the camera captures live stream data, which is then enhanced using Open Source Computer Vision Library (OpenCV) functions. These image enhancement techniques provide more detailed and clearer visuals from the webcam feed, making it easier to identify and track hand movements accurately. In real-world scenarios, camera feeds are often affected by various sources of noise and dynamic environmental factors such as lighting fluctuations, motion blur due to abrupt hand movements, and sudden changes in background. These factors can significantly affect the quality of visual input, making it difficult to reliably detect and track hand gestures. To address these challenges, the proposed system uses a series of image enhancement techniques, such as noise filtering, adaptive contrast enhancement, and motion blur compensation. Incorporation of these enhancement techniques improves the robustness and responsiveness of the vision-based control framework under real-time and dynamically changing conditions.

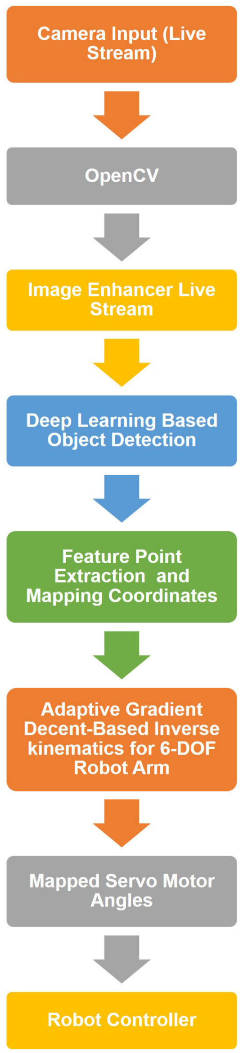


FIGURE 6. Flowchart of proposed vision-based control system.

Next, the enhanced live stream data is sent to deep learning based object detection using MediaPipe, a cross-platform framework for building multimodal applied ML pipelines. Within deep learning based object detection, hand key points are estimated using its advanced hand tracking model. This model identifies and tracks 21 landmarks on the hand, providing precise spatial information about each point's location.

Among these landmarks, we focus on landmark 8, which corresponds to the tip of the index finger. The x and y coordinates of this landmark are extracted, representing its pixel location within the video frame. To obtain depth information (the z-axis), we calculate the mean of the Euclidean distances from keypoint 9 (base of the middle finger) to keypoint 8 (tip of the index finger) and from keypoint 9 to keypoint 12 (tip of the middle finger). These distances were chosen after extensive testing, as it proved to be more stable compared to distances between other landmark pairs, offering more consistent and accurate depth estimation.

TABLE 3. Description of hardware used.

Hardware used	Description
Dexter ER2 Robotic Arm	The Dexter ER2 robot from Nex Robotics is a 5-axis robotic arm designed for educational and research purposes. With its modular structure, high-torque servo motors, and robust gripper, it offers precise manipulation capabilities. Widely used in academia, since Dexter provides a practical platform for learning kinematics, control algorithms manipulation tasks, the Dexter ER2 manipulator was selected [23]. It is equipped with heavy-duty servo motors (MG995). It delivers a strong turning force of almost 13 kg-cm through 180 degrees by its metal gears, and it is controlled by a simple Pulse Width Modulation (PWM) signal.
PCA9685 Servo driver	The PCA9685 is a servo driver module. Here, it allows control of multiple servos using a single Inter-Integrated Circuit (I2C) connection. It has 16 channels, each with 12-bit resolution for precise control. This means it can control the position of 16 servos independently. It communicates with microcontrollers like ATmega328P-based controllers or Raspberry Pi through I2C, freeing up General Purpose Input/Output (GPIO) pins for other uses
ATmega328P-based Controller	The ATmega-based Controller is the key component in vision-based tracking control of a 6-DOF robot arm, including integration of computer vision, vision-based control strategy, and enhanced performance.

After obtaining the x, y, and z values in pixel space, the data is mapped to real-world measurements in centimeters. This conversion is crucial to translate the visual information into physical dimensions that are compatible with our robotic arm's workspace. Also, a moving average filter is applied to the data. This filter helps to smooth out short-term fluctuations in the data and also highlights longer-term trends, effectively reducing noise in the hand tracking data without significantly reducing responsiveness.

The filtered x, y, z coordinates are then sent into an adaptive gradient descent-based IK solver for a 6-DOF robot arm. This solver takes the three-dimensional position as input and calculates the corresponding joint angles required for the robotic arm to reach that position.

VI. RESULTS AND DISCUSSION

This section presents the performance evaluation of the proposed system through both simulation and experimental studies. The simulation analysis of the proposed system is done in the Gazebo and PyBullet simulators. Further, the development of the experimental setup is presented, followed by the performance and robustness of the proposed system.

A. SOFTWARE AND HARDWARE USED

This section describes the hardware and software components used in the simulation and experimental studies. The hardware components of the experimental setup consist of a 6-DOF robotics arm, which isa Dexter ER2 Manipulator, an ATmega328P-based Controller, a personal computer (PC), and a servo driver.

The software components used include PyBullet simulator, Gazebo simulator, MediaPipe framework, OpenCV library, Robot Operating System 2 (ROS2), Arduino Integrated Development Environment (IDE), and for Computer-aided design (CAD) model development and Unified Robot Description Format (URDF) development, Fusion360 was used. The detailed description of hardware and software components is listed and described in Table 3 and Table 4, respectively.

TABLE 4. Description of software used.

Hardware used	Description
PyBullet Simulator	PyBullet is an open-source physics engine and robotics simulation toolkit for Python. It offers accurate simulations, robotics-focused tools, and ML integration. It's widely used by researchers to develop and test algorithms for motion planning, control, and other tasks before real-world deployment.
MediaPipe Framework	MediaPipe, an open-source framework [25], is used to do real-time perception tasks, including object detection and tracking. It is coupled with OpenCV to improve feature extraction for vision-based tracking control. Using OpenCV and MediaPipe, a system is created to recognize and track objects in real time, allowing for precise control of a 6-DOF robot arm.
OpenCV	OpenCV is an open-source machine vision library that detects the outlines and center of an item on a workstation. It provides several functions for machine vision algorithms, including contouring, Canny Edge, grayscale, and Gaussian filter.
Gazebo Simulator	Gazebo is an open-source 3D robot simulation environment. It offers various physics engines, realistic sensors, and a wide range of robot models. It also enables accurate testing of robotics algorithms in complex worlds that are customizable, which makes it a standard tool for development and validation in robotics research.
ROS2 (Humble Hawksbill)	Utilized as a primary middleware for robot control, communication, and simulation integration. It is used in conjunction with the Gazebo simulation environment to manage robot modeling, control interface, and data flow during development and testing.
Autodesk Fusion 360	Used for developing CAD models and generating URDF files for the robotic system.
Arduino IDE	Arduino IDE is a simple, cross-platform for programming Arduino boards. It has an intuitive interface, built-in libraries, and serial communication tools, enabling rapid prototyping and deployment of microcontroller-based projects.

B. SIMULATION RESULTS

The simulation and analysis in this work were conducted using two distinct approaches to validate the effectiveness of the proposed framework. The first utilized the Gazebo simulation environment integrating with the ROS2 Humble distribution for robotic modeling and control. In the second approach employed the PyBullet physics engine to simulate rigid body dynamics and interaction.

1) GAZEBO SIMULATION

In the first validation approach, which involves Gazebo-based simulation, the URDF file was developed to define the kinematic structure, link dimensions, and physical properties of the Dexter ER2 robotic manipulator. This URDF model was created using Fusion 360, a widely adopted CAD software that supports precise modeling and assembly of robotic components. Once the model was completed in Fusion 360, the URDF file was exported and subsequently imported into the Gazebo simulation environment. This integration enabled realistic visualization and simulation of the robot's behavior within the ROS2 framework, as illustrated in Figure 7.

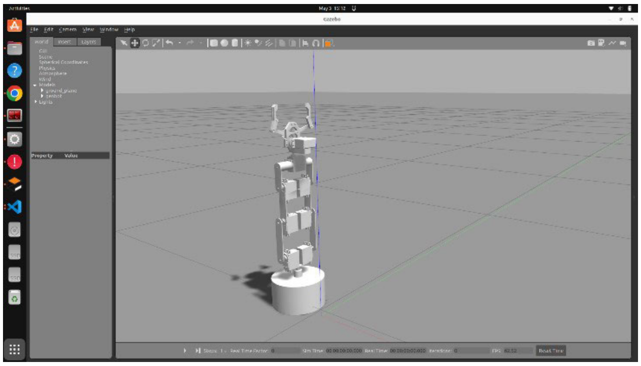


FIGURE 7. **Dexter ER2 spawned in Gazebo environment.**

To validate the effectiveness of the proposed IK solution, the adaptive gradient descent-based algorithm is implemented as a ROS2 Python node for computing the joint angles of the 6-DOF Robot Arm to achieve a desired end-effector position and orientation. This node is rigorously tested within the Gazebo simulation environment, where its performance is evaluated across various configurations using an iterative, trial-and-error approach. The integration of an IK solver with the Gazebo simulation enables real-time control and visualization of the robotic manipulator through ROS2 controllers, demonstrating the responsiveness and practical applicability of the proposed method in a simulated environment.

2) PyBullet SIMULATION

As a second validation strategy, the simulation and analysis approach utilized the PyBullet physics engine, which offers a robust and effective framework for simulating rigid body dynamics. In this setup, the proposed adaptive gradient descent-based IK solver is employed to compute the joint angles required to achieve a set of desired end-effector positions represented by their Cartesian coordinates (x, y, z). These computed joint angles were subsequently applied to update the configuration of the simulated manipulator within the PyBullet environment. The simulation results, illustrated in Figure 8, demonstrate the feasibility and effectiveness of the proposed control scheme under dynamic conditions.

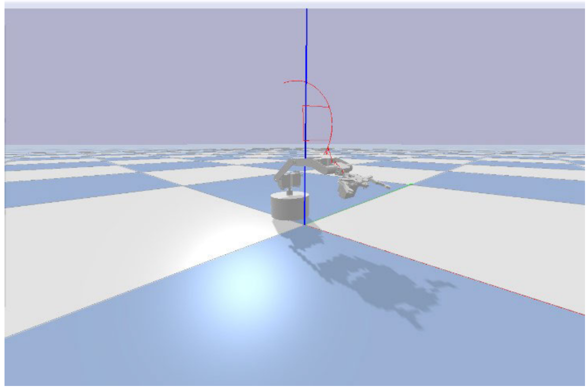


FIGURE 8. **PyBullet simulation.**

3) PICK-AND-PLACE TASK

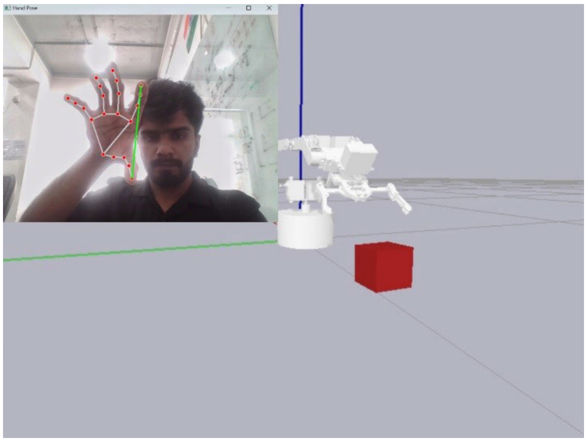
Figure 9 presents the execution of a pick-and-place task using the proposed gesture-driven control mechanism for the 6-DOF robotic arm. This task serve as a scenario-based validation to showcase the practical effectiveness and intuitive usability of the system. The results illustrate how real-time human hand gesture can be effectively translated into precise robotic manipulator actions, enabling accurate and efficient task completion. this emphasis the system's potential for deployment in real-world applications such as industrial automation, assistive robotics, and hazardous environment operations.

To evaluate the performance of the IK solver integrated into the control loop, the corresponding end-effector positions generated by the computed joint angles were compared with the initially provided desired positions. This comparison facilitates a detailed quantitative analysis of the IK algorithm's accuracy. The results were then plotted for data visualization, which is shown in Figure 10.

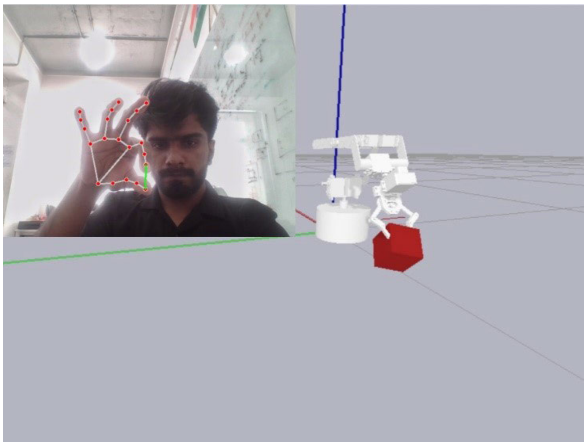
The analysis revealed a range of outcomes: in certain instances, the computed end-effector position derived from the desired coordinates by approximately 1 to 5 millimeters, while in others, near-exact alignment was achieved across the x, y, and z axes. These variations reflect the inherent challenges of solving IK for redundant and nonlinear systems, where factors such as numerical approximations, algorithmic convergence behavior, and proximity to kinematic singularities may influence accuracy. Despite these complexities, the overall results validate the reliability and precision of the adaptive gradient descent-based IK solver-based control scheme in a dynamic gesture-driven application context.

C. EXPERIMENTAL RESULTS

This section presents the experimental validation of the proposed system, which integrates a Dexter ER2 6-DOF



(a)



(b)

FIGURE 9. Pick and place simulations.

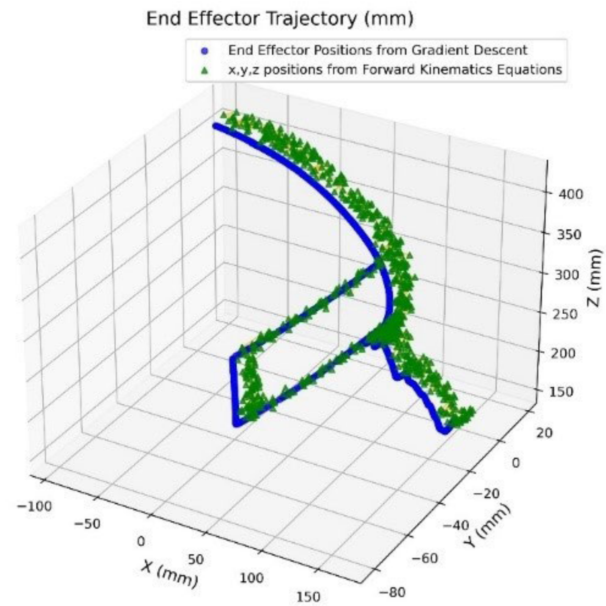


FIGURE 10. Data visualization to compare desired and actuated pose.

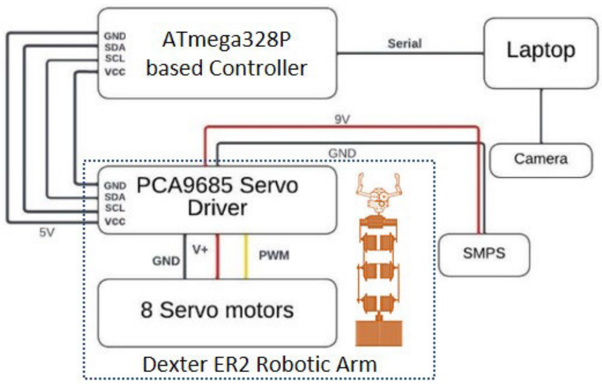


FIGURE 11. Hardware connection.

robotic arm with an ATmega328P-based controller. This integration transforms the setup into a dynamic system capable of tracking a moving hand using a monocular camera. In this configuration, the object of interest is a person's hand, detected through a monocular camera connected to a laptop. Operating initially in an idle state, the robotic arm mimics the hand's movements in real time.

The system utilizes a deep learning based gesture detection framework to extract real-time finger coordinates from a live video stream. These coordinates are processed using an adaptive gradient descent-based IK algorithm, tailored specifically for the robotic arm's kinematics. This allows the detection of hand gestures to be translated into corresponding joint angles, enabling the robotic arm to replicate the hand's pose and motion.

Figure 11 shows the hardware connection of an experimental setup. A PC with a control algorithm composed of the OpenCV library, MediaPipe, and ROS2 framework facilitates the integration of software tools with the Dexter ER2 Manipulator to achieve anthropomorphic control.

Figure 12 demonstrates the experimental validation of the proposed control scheme using a vision-based gesture-driven interface for the Dexter ER2 manipulator. Figure 12 (a) presents the experimental setup comprising the Dexter ER2 robot manipulator and the vision system used for hand tracking. In Figure 12 (b), the base joint of the manipulator is controlled by tracking the x-axis position of the hand in the image frame, with pixel coordinates translated into joint angles for positioning. Figure 12 (c) demonstrates the control of multiple joints by detecting the bend angle of the index finger, where the angle is mapped to coordinate joint movements of the manipulator. Figure 12 (d) shows the joint angle control using IK algorithm, taking x and y coordinates of the tip of index finger in the image frame, and z from the depth. Figure 12 (e) and (f) depict the gripper's response to hand gestures, where the opening and closing actions are triggered based on the relative distance between the thumb and wrist keypoints, facilitating intuitive grasp control.

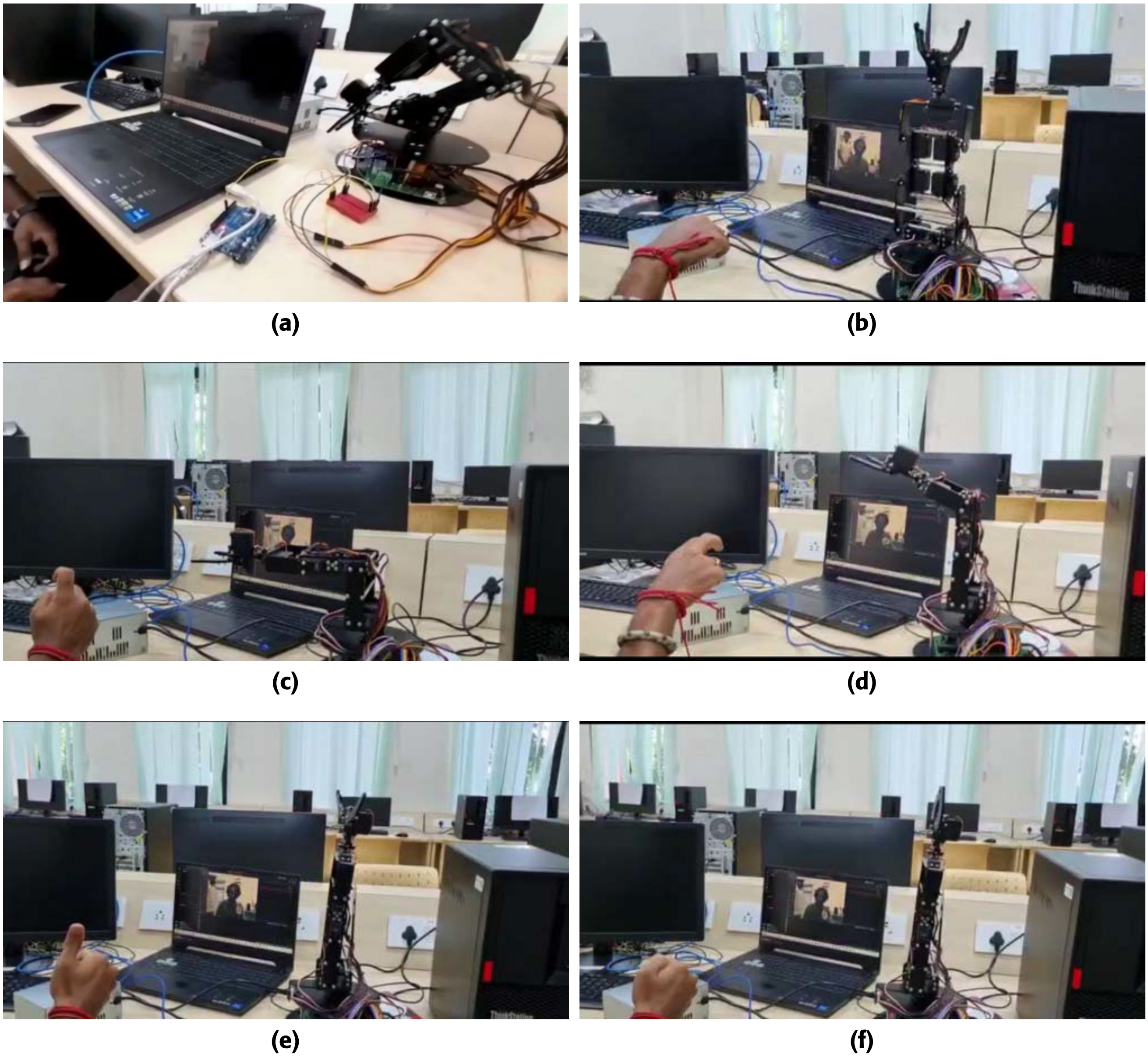


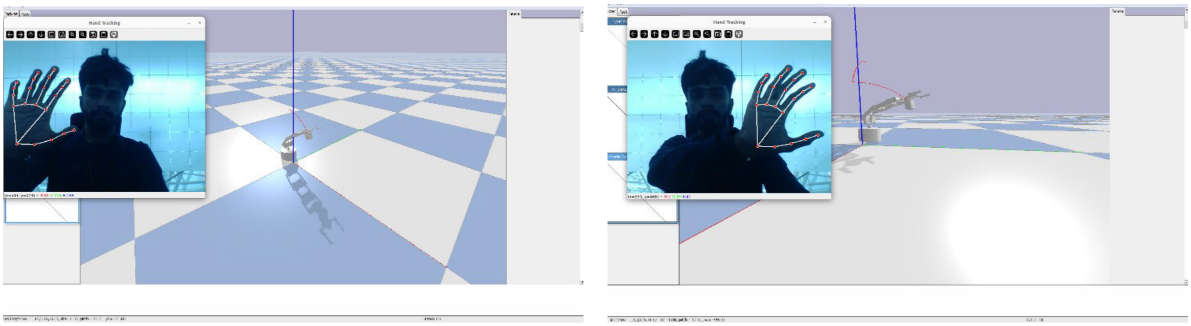
FIGURE 12. Experiment Result: (a) Experimental setup, (b) Control of the base joint rotation along x-axis of the image frame by converting pixel value into joint angles, (c) Controlling multiple joints by detecting the bend angle of the index finger and mapping this angle to corresponding manipulator joint movements, (d) Joint angle control using the IK algorithm, based on the fingertip's image position and depth information, (e) Gripper opens based on the distance between the thumb keypoint and the wrist keypoint, and (f) Gripper closes based on the distance between the thumb keypoint and the wrist keypoint.

VII. ROBUSTNESS TO NOISE AND DYNAMIC ENVIRONMENT

The robustness of the vision-based gesture detection components of the proposed control framework is validated through PyBullet-based simulation and real hardware setup, specifically under challenging conditions such as noisy input caused by lens distortion and low resolution camera, as well as dynamic environment factors including lighting variations and abrupt background changes.

As shown in inset window of Figures 13 (a) and 13 (b), the live stream is enhanced by applying noise reduction, adaptive contrast enhancement, and motion blur compensation techniques, as outlined in Section V. These enhancements significantly improve the reliability of gesture interpretation. Figures 13 (a) and 13 (b) illustrate the robotic arm's joint motion control, demonstrating its ability to follow the computed trajectories even in the presence of noise and environmental disturbance.

Inset window of Figure 12 further demonstrates the effectiveness ot the system in experimental conditions, where pixel values are successfully mapped between the camera frame and the robot's real workspace. Despite challenges



(a) (b)

FIGURE 13. PyBullet simulation: demonstrating robustness to noise and dynamic environment.

such as lens distortion, low-resolution input, and dynamic environmental factors like lighting variations and abrupt background changes, the system maintained its performance. This highlights the system's robustness and practical adaptability, underscoring the value of experimental validation in revealing real-world resilience beyond simulated conditions.

VIII. COMPARATIVE ANALYSIS

Table 5 presents the comprehensive comparative analysis of recent gesture-controlled robotic arm system presented in [8], [9], [12], [13], [14], [15], [16], [19], [21], and [22], by highlighting key aspects such as sensor used, hardware platform, AI model, system limitation, relative cost and remark.

The following summary of Table 5 highlights the novelty and effectiveness of the proposed approach.

- Many existing approaches are limited to simulation and lack real-time control capabilities, as seen in [8], [15], [21], and [22]. In contrast, the proposed work demonstrates real-time control through an experimental setup, bridging the gap between theory and practical implementation.
- In [16], a sensor fusion technique is employed, which involves complex and expensive hardware configuration. The proposed approach simplifies this by utilizing a single RGB camera, offering a cost-effective and easy-to-deploy solution without affecting functionality.
- Few prior studies rely on wearable devices for gesture-based control [9], [16], which may hinder user comfort and increase system complexity. The proposed method eliminates this dependency by using a vision-only system that interprets hand gestures without any physical attachment.
- Works like [8] and [13] are constrained by low precision and limited DOF. To address this, the proposed system employs a 6-DOF robotic arm coupled with an adaptive gradient descent-based IK solver, ensuring responsive control.
- Another challenge in [15], [19], and [21] is their susceptibility to noise and poor adaptability to environmental variations. The proposed model addresses these issues through a deep learning-based gesture recognition model that is both noise-resilient and performs under diverse environmental variations, ensuring robust performance in real-world environments.

This comparison clearly highlights the practical advantages of the proposed work in terms on intuitive control, low-cost implementation, robust performance in a noisy and dynamic environment, and real-time validation through both simulation and physical experimentation.

IX. DISCUSSION

This study explored the development and implementation of a gesture-based control interface for the Dexter ER2 robotic arm. The primary aim was to create an intuitive system that allows users to control the robotic arm through natural hand movements, thereby reducing the cognitive load typically associated with traditional robotic control methods. This discussion synthesizes the key threads from our research, highlighting the implications of our findings.

A. OPEN RESEARCH QUESTIONS AND FUTURE DIRECTIONS

The Open Research Questions (ORQs) identified from the present study and related research include enhancing the robustness and accuracy of gesture-driven control while keeping the system affordable. This study addresses the intuitive control in robotic systems using gesture-driven control, which is the central theme of the present research. Traditional methods, such as joysticks and keyboards, often require extensive training and can be counter-intuitive, particularly for novice users. Proposed gesture-driven approach leverages natural human movements, making it easier for users to learn and operate the robotic arm effectively. This aligns with existing literature that emphasizes the need for user-friendly interfaces in robotics to enhance accessibility and usability. By employing gesture-driven control, the aim is to minimize cognitive load during operation while

TABLE 5. Comparative analysis.

S.No	Author(s), Year [Reference]	Sensor used	Hardware Used	Model Used	Type of Robot	Limitation	Relative Cost	Simulation/ Experiment	Remarks
1	Atre *et al.*, 2018 [8]	Webcam	Arduino, Bluetooth module	None (Rule-based)	Robotic arm in simulator	Low precision, limited DOF	Low	Simulation	Limited to hand gesture, Lacks AI integration, no real-time control of robotic arm
2	Angulo *et al.*, 2022 [12]	Stereo vision camera	Custom-built robot arm	Classical vision techniques	Robotic arm	Requires 3D stereo vision, not intuitive	High	Experiment	Complex setup, may not suitable for general task
3	Solly and Aldabbagh, 2023 [9]	Flex sensor, Bluetooth module, accelerometer, radio module	Wearable hand glove controller system	Gesture map	Mobile robot	Wearable, not suitable for precise manipulation	High	Experiment	Focused on mobility, wearable
4	Altayeb, 2023 [13]	Webcam	Arduino nano, servo motors, buck converter	None (simple vision thresholding)	Fingers of robot arm	Low scalability	Moderate	Experiment	Basic hand detection, lacks adaptive control
5	Qin *et al.*, 2023 [14]	Camera (vision only)	Teleoperation server	Imitation learning algorithm	Multi-fingered dexterous robots	Loss of tracking during fast motion, unreliable hand pose when	High	Both Simulation and Experiment	Lacks IK for pose estimation
6	Suhaimi *et al.*, 2023 [16]	Electrooculogram (EOG), EMG sensor	4-DOF robotic arm	EMG gaze estimation and classical image processing	4-DOF robotic arm	Complex sensor fusion, expensive	Very High	Experiment	Too complex and expensive for practical use
7	Terreran *et al.*, 2023 [15]	RGB-D camera	None (Simulation study)	Ensemble learning based gesture recognition	Only gesture recognition	No real-time control of robot	Low	Simulation	Sensitive to noise and environmental condition so not suitable for complex task.
8	Bamani *et al.*, 2024 [19]	RGB web-camera	Unitree Go1 quadruped robot	Super-resolution model based gesture recognition	Quadruped robot	Limited gesture, challenging in variable environmental conditions	Very High	Experiment	Limited set of gesture, faces challenges under varying environmental conditions.
9	Assis *et al.*, 2025 [22]	RGB camera	None (Simulation study)	Sequential ML model	6-DOF robotic arm in simulator	No real-time testing mentioned	Low	Simulation	Real-time implementation is not addressed
10	Ferin, 2025 [21]	RGB Camera	None (Simulation study)	deep learning based gesture recognition	Only gesture recognition	No real-time testing on robotic arm mentioned	Low	Simulation	Sensitive to lighting conditions and occlusions, Real-time robotic control not demonstrated
11	Proposed Work	RGB web-camera	Dexter ER2 (6-DOF arm), and servo driver	deep learning based gesture recognition and adaptive gradient decent-based IK solver	6-DOF Robotic Arm	Sensitivity to camera-to-person distance	Low	Both Simulation and Experiment	Vision-Based, No wearable, Adaptive Gradient Decent IK solver, practical, robustness to noise and dynamic condition

improving the robustness with noise and environmental variations. The proposed scheme demonstrated that intuitive gesture recognition could facilitate smoother interactions with the robotic arm, allowing users to focus more on task execution rather than on mastering intricate control mechanisms. This finding has significant implications for designing future robotic interfaces and enabling seamless adaptation to smarter manufacturing environments aligned with Industry 4.0 technologies. Another ORQ addressed by the present research was maintaining affordability while achieving effective control over the robotic arm. Further performance

The design considerations ensured that the system could be implemented without significant financial burdens, making it accessible to a broader user base, including educational institutions and small businesses. This focus on affordability is essential in democratizing access to advanced robotics technology.

Further ORQs are latency issues in real-time control systems, integrating multi-modal control approaches (e.g., combining gesture and voice inputs), and enabling seamless adaptation to smarter manufacturing environments aligned with Industry 4.0 technologies that can be addressed in future work. Also, another ORQ is the impact of camera-to-person distance on system performance, which may affect human-robot interaction. This limitation can be addressed in future work by incorporating a super-resolution model-based gesture recognition to enhance the accuracy at varying distances. Also, these ORQs can be addressed by exploring advanced ML techniques to improve gesture recognition in dynamic environments. Further latency in real-time robotic control can be minimized by investigating optimized communication protocols. Also, modular, scalable frameworks can be developed that support personalized user interfaces for industrial robots.

B. PRACTICAL MANAGERIAL SIGNIFICANCE

This applied research emphasizes the practical managerial significance of implementing an anthropomorphic control system for a compact 6-DOF robotic manipulator. By utilizing a gradient descent algorithm with multiple learning rates for IK and pose estimation via a single monocular camera, the system offers a cost-effective, adaptable solution for remote operations. Its intuitive design and natural control interface empower managers to deploy robotics effectively in challenging environments, prioritizing usability over high precision.

The adaptable gradient descent-based IK enables efficient positioning of the manipulator's end-effector in both simulation and real-world settings, reducing the need for complex re-calibrations during deployment. The anthropomorphic control approach provides a user-friendly interface, enhancing operational efficiency by allowing personnel to intuitively control the manipulator, particularly in hazardous environments such as nuclear facilities, disaster zones, or chemical plants.

The reliance on a single-camera setup for pose estimation minimizes hardware complexity and cost, making this system particularly suitable for resource-constrained scenarios, like small businesses or organizations operating in remote areas. Additionally, the integration of simulation tools like Gazebo and PyBullet ensures robust pre-deployment testing and validation, enabling decision-makers to mitigate risks and optimize performance before full-scale implementation.

By combining intuitive, human-centric control mechanisms with computational efficiency, this research demonstrates the value of robotics systems in addressing critical operational challenges in inaccessible or hazardous environments. It highlights the need for adaptable algorithms and user-friendly designs to enhance the practical utility and scalability of robotic manipulators, ultimately offering a reliable, scalable tool to improve safety, efficiency, and operational reach.

X. CONCLUSION

This work presents a vision-based gesture-driven control of a 6-DOF robotic arm using a monocular camera, offering a cost-effective and less computationally intensive control algorithm. The proposed method integrates deep learning-based hand feature detection with an efficient IK solver to enable intuitive, real-time control. The feature-based gesture detection algorithm processes input image data, while the MediaPipe framework utilizes deep learning to extract visual features from a human's moving hand. The system's performance was validated through simulations and real-world experiments, demonstrating its potential for accurate and responsive robotic manipulation. Robustness and comparative analysis further highlight the advantages of the proposed scheme. Minor deviation ranging from 1 to 5 mm, with near-exact alignment in some cases, highlight the challenges of IK in redundant systems, while validating the reliability of the adaptive gradient descent-based solver in gesture-driven control.

As a future scope, the system can be further validated through a comprehensive experiment on a real-world task to ensure reliability and practical applicability.

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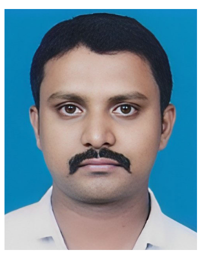

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